

ROV Drop-Weight Emergency Release

Burn-wire (nichrome/nylon) design — fully external, fully flooded, zero moving parts through the hull. Companion to the NEPTUNE camera, navigation and blackbox specs.

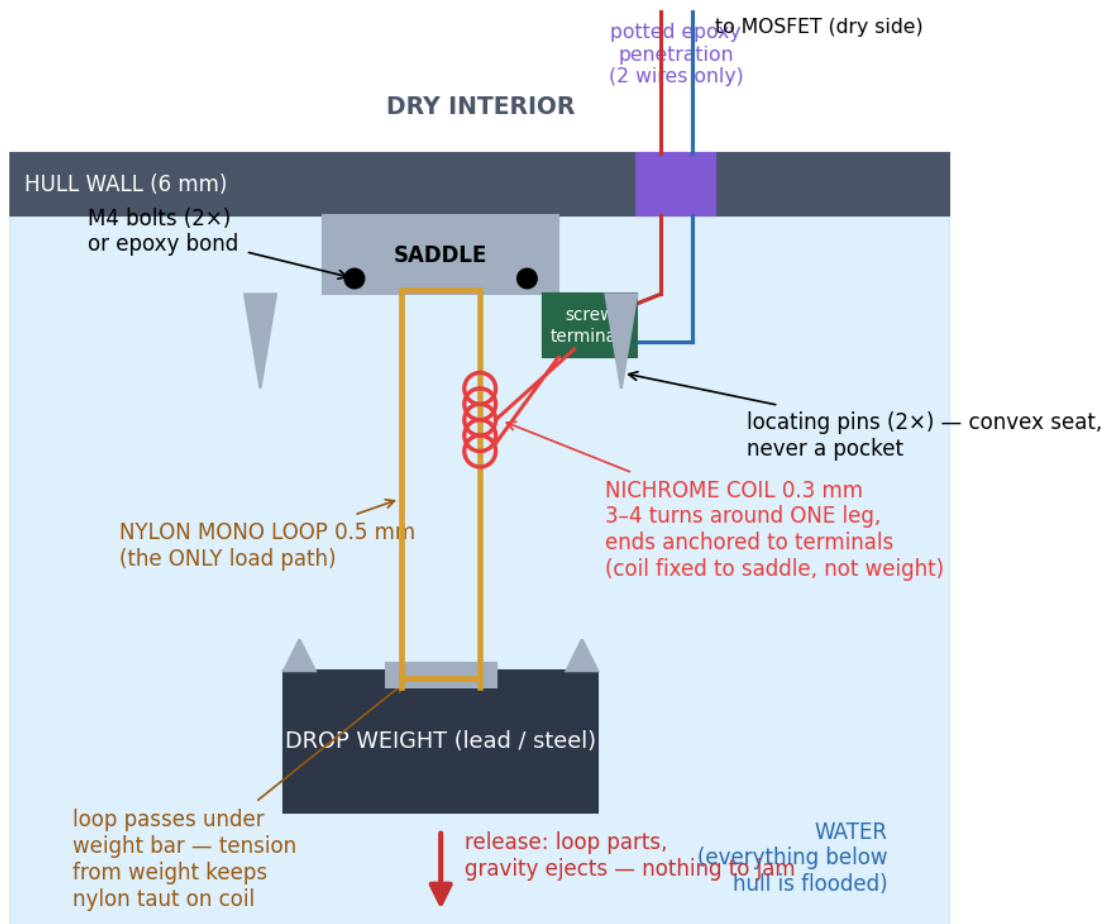
1 · The design principle

The problem you were trying to solve — a trigger mechanism that deploys without leaking — dissolves once the whole mechanism moves **outside** the hull. Every oceanographic release works this way. Nothing crosses the pressure boundary except two ordinary wires in a potted epoxy penetration, identical to the pass-throughs you already run for thrusters and sensors. There is no shaft seal, no dry pocket, no part that can flood, and therefore no implosion path.

The second principle: **the weight never hangs on the burn wire**. A nylon monofilament loop is the only load path; the nichrome coil merely wraps one leg of that loop and melts it on command. Holding and releasing are separated, which is what makes the mechanism simple, reliable, and cheap to re-arm.

Burn wire works *better* under tension and works fine fully submerged: the coil only needs to reach ~200 °C locally for an instant. Water roughly doubles the current requirement, which is why the design uses 0.3 mm nichrome fired at 2–3 A rather than thinner wire at lower current.

FIG 1 — Side cross-section: complete release assembly (flooded, external)

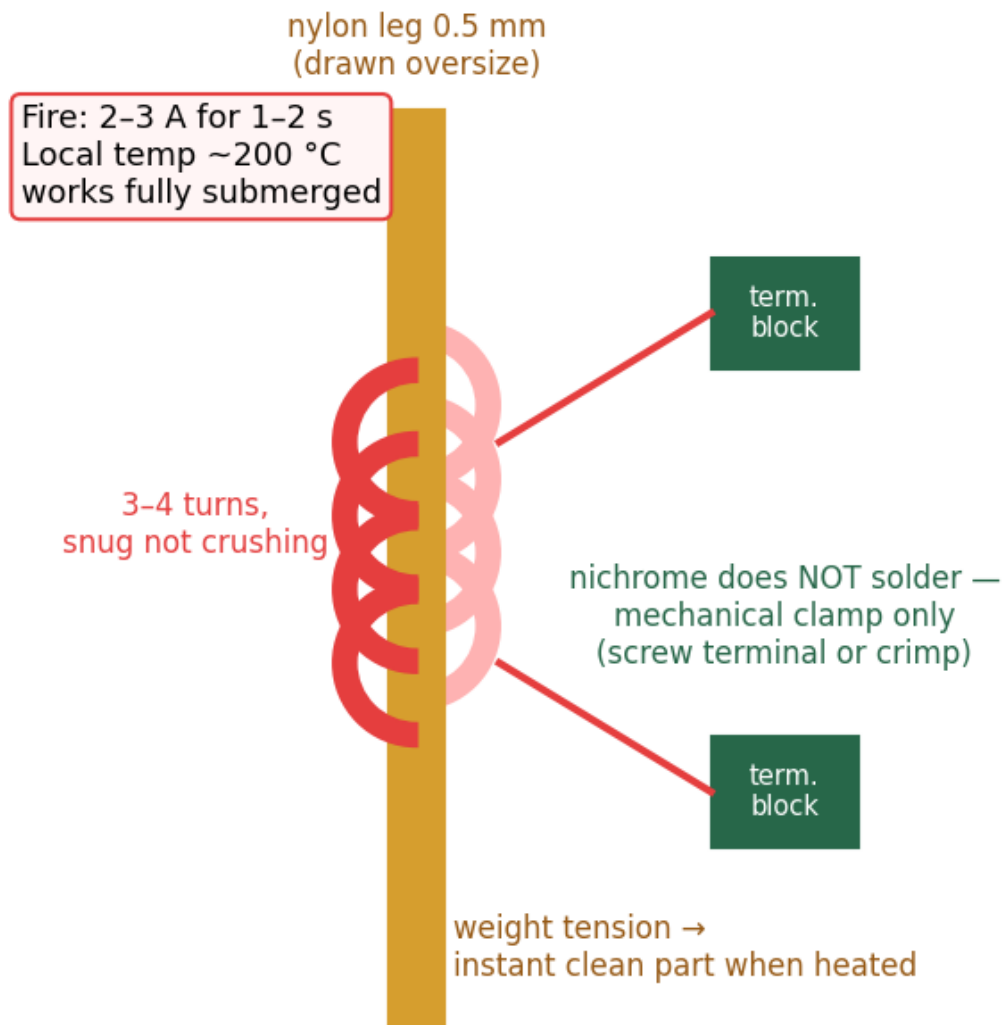


2 · The release interface

3–4 snug turns of 0.3 mm (28 AWG) nichrome around one leg of the 0.5 mm nylon loop. Both nichrome tails anchor to a small screw-terminal block on the saddle — nichrome cannot be soldered, so every joint is mechanical: screw terminal, crimp, or wrap-and-washer. The coil is fixed to the *saddle*, never to the weight, so the falling weight cannot drag the coil or its wiring with it.

Tension matters. The weight's own mass keeps the nylon taut against the coil, and tensioned mono parts instantly and cleanly when heated. Slack line melts slowly and can re-fuse. If you test the mechanism unloaded on the bench, it will look worse than it behaves in service.

FIG 2 — Detail: nichrome-on-nylon interface

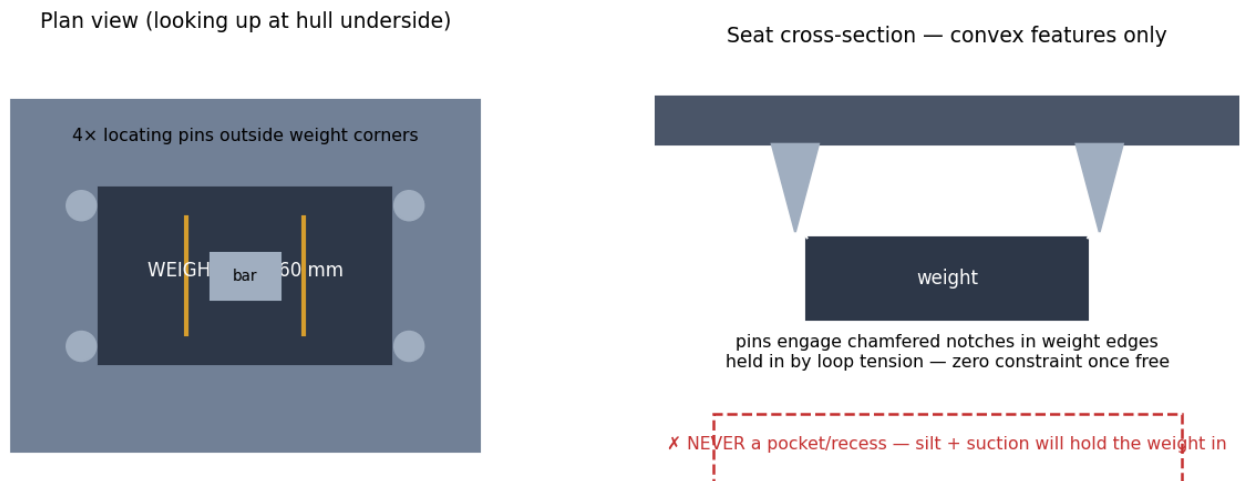


3 · Seating the weight — no jiggle, no jam

The seat constrains the weight in every axis while held, and constrains nothing once the loop parts. Four locating pins (or a shallow V-channel) engage chamfered notches on the weight's edges; the taut loop pulls the weight up into that seat. Gravity is the entire ejection mechanism — the weight hangs over open water with nothing below or around it.

The one geometry to refuse: a pocket or recess the weight sits *inside*. Canal silt packs into pockets, and a flat weight in a flat-bottomed recess adds suction. Convex features only — pins, ridges, chamfers. If the weight can be pushed sideways out of its seat when the loop is cut, it can also fall out; that is the property you want.

FIG 3 — Weight seating: constrained while held, free when released

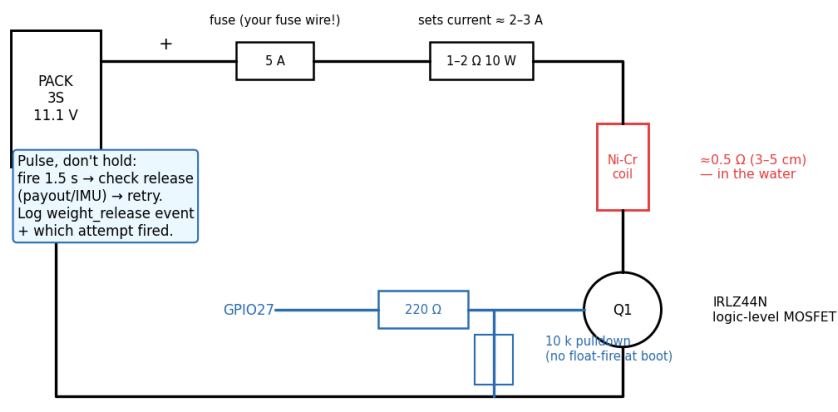


4 · Firing circuit

The nichrome coil is $\sim 0.5 \Omega$, near enough a short circuit, so the series power resistor is what sets the firing current — $1\text{--}2 \Omega / 10 \text{ W}$ caps a 3S pack at roughly $2\text{--}3 \text{ A}$ wet-rated firing current. The MOSFET is a logic-level IRLZ44N-class part driven from a GPIO through 220Ω , with a 10 k pulldown so a floating pin at boot can never fire the release. Re-use the fuse-wire card as an actual 5 A fuse in this line: correct rating, zero cost, and it protects against a shorted penetration.

Fire in pulses, never held: 1.5 s on, check for release (tether payout jump, IMU pitch change, depth rate), retry if needed. Log every attempt to the blackbox as a `weight_release` event with the attempt number and which path fired. Expect a visible pack-voltage dip in the power trace during the pulse — that dip doubles as free confirmation the coil actually drew current.

FIG 4 — Firing circuit (dry side) + failsafe hierarchy



FAILSAFE HIERARCHY: 1° electromagnet — holds weight seated; POWER LOSS = DROP (passive). 2° burn wire — commanded backup, separate GPIO + separate penetration.

5 · Failsafe hierarchy

The burn wire is the **commanded** release. The **passive** release remains the electromagnet: it holds the weight seated whenever the system is powered, so total power loss drops the weight with no computation and no command. The two paths use different physics, different penetrations, and different GPIO channels, so no single fault disables both. Arming order at the bank: seat weight → tie loop → energise magnet. Surfacing normally: carry the weight home. Emergency: magnet off (or power lost) drops it; burn wire severs the loop if the magnet path has failed mechanically.

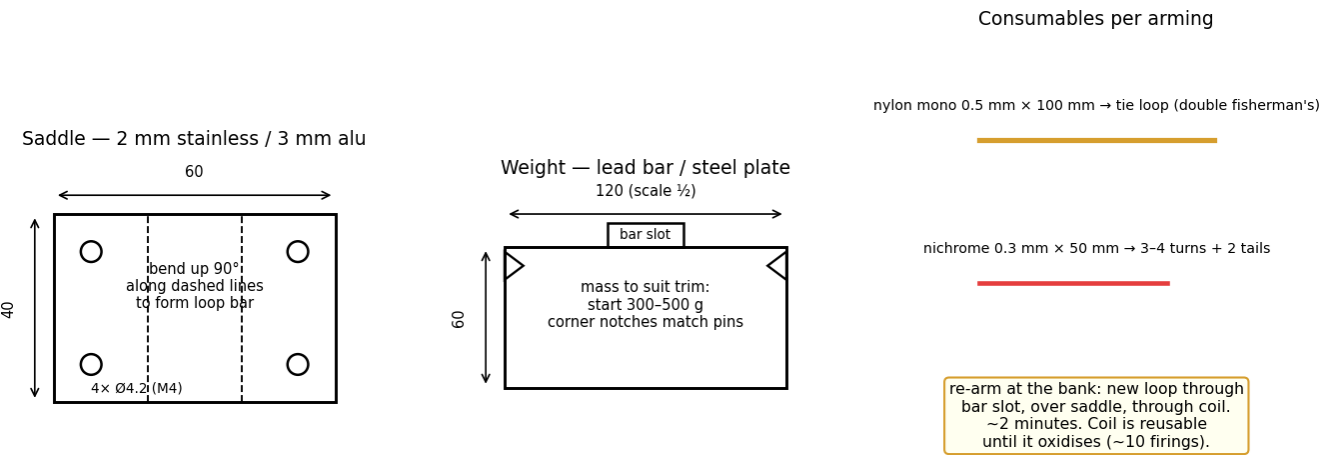
6 · Bill of materials

Item	Spec	Qty	~Cost	Notes
Nichrome wire	0.3 mm / 28 AWG (Kanthal fine)	1 spool	£2	vape shop; ~10 firings per coil
Nylon monofilament	0.5 mm fishing line	1 spool	£2	1 loop = 100 mm, consumable
Power resistor	1–2 Ω, 10 W wirewound	1	£1	sets firing current
MOSFET	IRLZ44N (logic-level)	1	£1	conducts ~2 s per firing, no heatsink
Resistors	220 Ω gate, 10 kΩ pulldown	2	£0.10	
Fuse	5 A (the fuse-wire card)	1	£0	inline protection
Saddle	2 mm stainless or 3 mm alu strip	1	£2	cut/bend per FIG 5
Terminal block	2-way screw terminal, small	1	£0.50	nichrome anchor
Locating pins	M4 stainless bolts or nylon standoffs	4	£1	
Weight	lead bar / steel plate 300–500 g	1	£0–3	mass to suit trim
Epoxy	potting for penetration + magnet coat	—	on hand	

Total new parts: roughly **£10**.

7 · Cut & drill templates

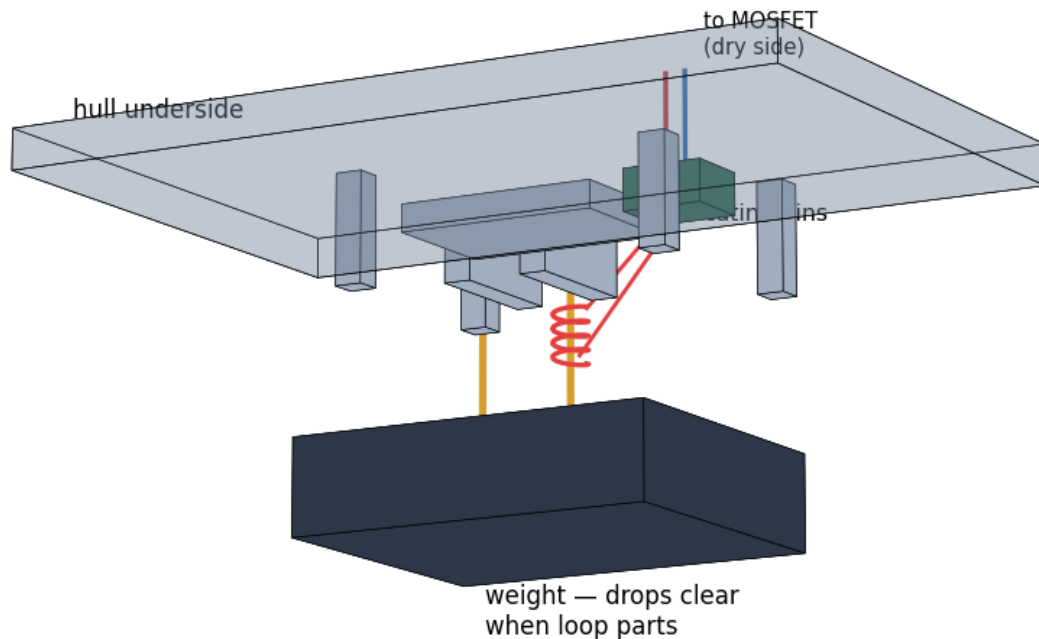
FIG 5 — Cut & drill templates (mm)



Dimensions are a sane starting point, not gospel — scale the saddle and weight to your hull. The invariants: loop legs 10–20 mm apart, coil positioned mid-leg with ≥15 mm clearance from both saddle and weight, pins clear of the weight by ~1 mm per side when seated.

8 · 3D assembly view

FIG 6 — 3D assembly view (exploded slightly for clarity)



9 · Assembly sequence

1. Cut and bend the saddle (FIG 5). Drill 4× Ø4.2 mounting holes and fit the terminal block.
2. Bolt or epoxy the saddle to the hull underside. Fit the 4 locating pins around the weight footprint.
3. Pot the 2-wire penetration through the hull (same epoxy technique as the thruster wires). Wire the dry side per FIG 4: fuse → power resistor → penetration; MOSFET low-side; gate 220 Ω to GPIO27; 10 k pulldown.
4. Wind the nichrome coil: 50 mm of 0.3 mm wire, 3–4 turns around a 0.5 mm drill bit as a former, two straight tails. Clamp both tails in the terminal block so the coil sits mid-air where the loop leg will run.
5. Cut 100 mm of nylon mono. Thread: under the weight bar, up both sides, one leg through the coil, tie over the saddle bar with a double fisherman's knot. Trim tails.
6. Lift the weight into its seat; the loop should go taut and the weight should not rattle when shaken.
7. Bench-fire in a bucket (see §10) before first dive.

10 · Test procedure

Seat test (dry): weight seated, loop tied. Shake the hull in all axes — no rattle, no rotation, no visible movement. Push the weight sideways firmly: it must not walk out of the pins while the loop is intact.

Release test (bucket): submerge the assembly, fire a 1.5 s pulse from the dashboard. Weight must be gone in under 2 s. Record the measured current on a bench supply the first time — that number goes in the config as the expected firing current, and the blackbox power trace is checked against it on every real firing.

Repeat ×5 with fresh loops before trusting it on a dive. Then re-test once a season and after any silt-heavy outing. Replace the coil when it looks oxidised/flaky — roughly every 10 firings.

Dashboard integration: the release button lives behind an explicit confirm (like SD format), fires the pulse-check-retry sequence automatically, and logs `weight_release` with attempt count, measured pack sag, and outcome. The SURFACE control should offer it as the escalation step if ballast blow fails.

11 • Why not the alternatives

Fuse wire as the heater: copper's milliohm resistance barely warms at sensible currents and conducts heat away from the nylon; pushing harder fuses the wire before the nylon melts. **Fuse wire as a fusible link:** workable dry (blow the wire itself at 10–20 A through a limiting resistor), but it demands a dry pocket — a seal you then own forever. **2 mm nichrome:** needs 40–60 A to glow; thicker is harder to heat, not easier. **Servo/solenoid latch:** a moving part crossing the boundary or living in silt — the exact failure modes this design deletes. The nichrome-on-nylon release has no seals, no moving parts, £10 of parts, and two minutes to re-arm.